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EU AI Act

AI Regulatory Sandboxes

Best Practices for AI Stakeholders

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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We’ve seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it’s handled responsibly. That’s why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

What Do We Do?

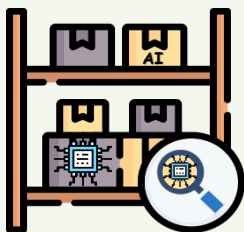
We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.



How Do We Do It?

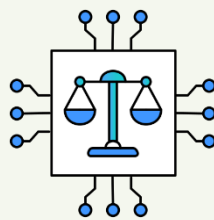
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building ‘Trustworthy-by-Design’.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.



Contents

Who Are We	2
Introduction	5
Key questions being asked about AI Regulatory Sandboxes	6
1. What are AI regulatory sandboxes and how do they differ from blockchain sandboxes?	7
2. Why should AI regulatory sandboxes look to blockchain sandboxes for inspiration?	7
3. How have blockchain sandboxes dealt with regulatory fragmentation across Member States?	7
4. What lessons can be learned regarding regulator engagement and onboarding?	7
5. How can AI sandboxes handle multi-regulatory compliance like GDPR, eIDAS, and DSA?	7
6. How did blockchain sandboxes manage open-source use cases and responsibilities?	8
7. What technical documentation practices can AI sandboxes adopt from blockchain ones?	8
8. How can AI sandboxes structure effective dialogue rounds?	8
9. What role does pan-European coordination play in sandbox success?	8
10. What should AI sandboxes avoid based on blockchain sandbox experiences?	8
Understanding AI Regulatory Sandboxes	9
1. Introduction (Cohort Overview)	10
2. Terminology, Laws, and Regulatory Framework	11
3. Data Protection (GDPR)	12
4. Relevance of eIDAS and Trust in AI Systems	13
5. AI & Enabling Technologies (e.g. IoT)	14
6. Legal Entity Structures for AI Model Providers	15
7. AI in Democratic Processes	16
8. AI for ESG and Sustainability Reporting	17
9. AI in Supply Chain Due-Diligence	18
10. Product Passports, Smart Labelling, and Digital Twins	19
11. Digital Governance Act & Digital Services Act	20
12. AI in Sanctions Screening & Compliance Automation	21
13. AI and Anti-Money Laundering (AML)	22
14. Digital Operational Resilience (DORA) for AI	23
15. Classification of AI Systems under EU AI Act	24
16. Real-World Testing and Sandbox Exemptions	25
17. Open Source AI – Roles & Responsibilities	26
18. AI for Crowdfunding, Finance, and Tokenised Assets	27
19. Automated Decision-Making & Algorithmic Markets	28
Mapping AI Regulatory Sandboxes to Ecosystem Actors	29





Calls to action	37
Conclusion	39
Appendix A: A Practitioner's View: Navigating the 'Triple Bind' of Multi-Regulatory Compliance in AI Sandboxes for Financial Services	40
Appendix B: A Practitioner's View: Rights-Respecting Approach to Regulation	42
References.....	43





Introduction

As the EU Artificial Intelligence Act (AI Act) moves toward implementation, regulatory sandboxes have emerged as a key mechanism for supporting safe, transparent, and innovative AI development. These supervised environments enable AI systems—particularly those classified as high-risk—to be tested under real-world conditions while benefiting from regulatory guidance and controlled flexibility.

This report serves as a practical blueprint for AI firms, regulators, and users navigating the design and use of AI regulatory sandboxes. It distils best practices, lessons learned from adjacent frameworks (such as blockchain sandboxes), and actionable insights for harmonizing sandbox implementation across the EU. From risk classification and legal interoperability to user rights and post-market monitoring, the report identifies opportunities to align sandbox activities with the AI Act's requirements and broader digital governance goals.

The sandbox initiative represents a crucial step in the EU's efforts to develop a robust, forward-looking AI governance framework. It aligns with the EU's vision to balance innovation and accountability under the AI Act by providing a controlled environment to test AI systems while ensuring adherence to legal requirements. The sandbox supports the twin goals of fostering innovation and maintaining public trust, ultimately promoting the safe and ethical deployment of AI technologies.

In mapping out key roles, responsibilities, and interactions among sandbox participants, the document promotes a collaborative, rights-respecting approach to AI experimentation. It highlights how firms can accelerate compliance, how regulators can build technical capacity, and how users can benefit from more trustworthy and explainable AI systems before they reach the market.

Whether you are a startup testing a foundation model, a supervisory authority shaping enforcement strategies, or a civil society group monitoring algorithmic impacts, this report provides a structured reference for engaging with AI regulatory sandboxes—laying the groundwork for responsible, scalable, and lawful innovation in the EU AI ecosystem.

Best regards,

Sean Musch

Founder/CEO

AI & Partners



Key questions being asked about AI Regulatory Sandboxes





1. What are AI regulatory sandboxes and how do they differ from blockchain sandboxes?

AI regulatory sandboxes are controlled environments established under the EU AI Act to test and validate innovative AI systems under regulatory oversight. They differ from blockchain sandboxes, such as the European Blockchain Sandbox, in that they focus on AI-specific risk classifications, transparency, and human oversight obligations. However, blockchain sandboxes already offer experience with cross-border coordination, layered supervision, and technological convergence (e.g., DLT + AI), which are highly relevant for AI. Learning from blockchain pilots can prevent duplication and foster aligned regulatory expectations.

2. Why should AI regulatory sandboxes look to blockchain sandboxes for inspiration?

Blockchain sandboxes have developed proven protocols for engaging multi-sectoral regulators, managing cross-border use cases, and facilitating iterative compliance. They offer concrete best practices in stakeholder coordination, role clarification, and legal boundary-setting—areas equally complex in AI. By leveraging these lessons, AI sandboxes can avoid early-stage confusion, structure dialogues more effectively, and build trust among innovators and regulators. Blockchain sandboxes also offer tested formats for handling overlapping legislation (e.g., GDPR + sectoral laws), which mirrors the AI landscape under the AI Act, DSA, and Data Act.

3. How have blockchain sandboxes dealt with regulatory fragmentation across Member States?

European Blockchain Sandboxes use coordinated dialogues and regulator mapping to overcome fragmentation. Each use case is matched with multiple national and EU authorities, ensuring holistic assessment. For AI, where national authorities must implement Article 57–59 of the AI Act, similar mechanisms are critical. By mirroring this model, AI sandboxes can pre-empt divergences in how high-risk classification, documentation, and sandbox eligibility are interpreted, promoting harmonization. Cross-border matchmaking also builds trust between regulators and improves interoperability of national sandbox programs.

4. What lessons can be learned regarding regulator engagement and onboarding?

Blockchain sandboxes successfully implemented pre-dialogue briefings, onboarding webinars, and blockchain expert sessions to bring regulators to a shared knowledge baseline. AI sandboxes can adopt similar techniques, including sector-specific AI training, pre-briefings on use case contexts, and regulator-only coordination sessions between rounds. These mechanisms ensure that dialogue is constructive and focused, avoiding delays due to uneven technical understanding or regulatory uncertainty. Clear preparation tools lead to more actionable feedback for innovators and better internal coordination among supervisory authorities.

5. How can AI sandboxes handle multi-regulatory compliance like GDPR, eIDAS, and DSA?

Blockchain sandbox dialogues often addressed dual or triple compliance regimes (e.g., GDPR + eIDAS + AML). They developed templates and checklists to help innovators map compliance pathways across regulatory domains. AI sandboxes, facing similar overlaps (e.g., AI Act + GDPR + sectoral finance or health laws), should adopt modular legal mapping tools and foster joint regulator engagement in dialogues. This helps innovators understand composite obligations and reduces the risk of inconsistent advice from separate regulators.





6. How did blockchain sandboxes manage open-source use cases and responsibilities?

Blockchain sandboxes encountered complex questions about liability, traceability, and obligations in open-source DLT solutions. They addressed these through role-mapping exercises, shared legal interpretations, and clearer delineation of controller-processor responsibilities. These methods are directly applicable to AI, especially for foundation models and open-source AI tools. AI sandboxes should build on these techniques to guide innovators in defining the limits of their obligations under Article 25(4) of the AI Act, and in clarifying the role of third-party contributors.

7. What technical documentation practices can AI sandboxes adopt from blockchain ones?

Blockchain sandboxes promoted best practices like using standard templates for smart contract documentation, system diagrams, and data flows. AI sandboxes can build on this by encouraging the use of structured model cards, data sheets, and system impact assessments (SIAs). Consistent documentation enables regulators to understand AI behaviour, supports transparency under the AI Act, and aligns with the documentation duties for high-risk systems. Requiring early submission of structured artifacts can streamline sandbox dialogue and improve feedback loops.

8. How can AI sandboxes structure effective dialogue rounds?

Blockchain sandboxes implemented a dual-meeting format: an initial dialogue for presentation and Q&A, followed by a second, deeper session after regulator-only meetings. This format improved feedback quality and allowed regulators to coordinate before re-engaging with the innovator. AI sandboxes should replicate this structure to enhance clarity, surface conflicts in interpretation, and allow for thoughtful regulator alignment. Adding a final wrap-up session with post-dialogue guidance would further improve the utility of participation for both firms and authorities.

9. What role does pan-European coordination play in sandbox success?

One of the blockchain sandbox's key achievements was facilitating regulatory convergence through shared learnings and coordinated dialogues. This pan-European perspective ensured alignment across jurisdictions and avoided market fragmentation. AI sandboxes must likewise function beyond national borders, especially since many AI solutions are deployed at EU scale. Coordination via networks like the European Forum for Innovation Facilitators (EFIF) and shared platforms for sandbox learnings can build a consistent supervisory ecosystem for AI across the EU.

10. What should AI sandboxes avoid based on blockchain sandbox experiences?

Blockchain sandboxes initially struggled with vague role definitions, late regulator engagement, and lack of follow-up from earlier cohorts. AI sandboxes should avoid these pitfalls by establishing clear governance protocols, documenting lessons transparently, and ensuring each round builds upon prior findings. Additionally, sandboxes must resist over-legalization—flexibility and mutual learning should remain central. Creating rigid entry criteria or overly prescriptive outputs may deter innovators. Instead, fostering trust, transparency, and responsiveness will yield more impactful and scalable results.

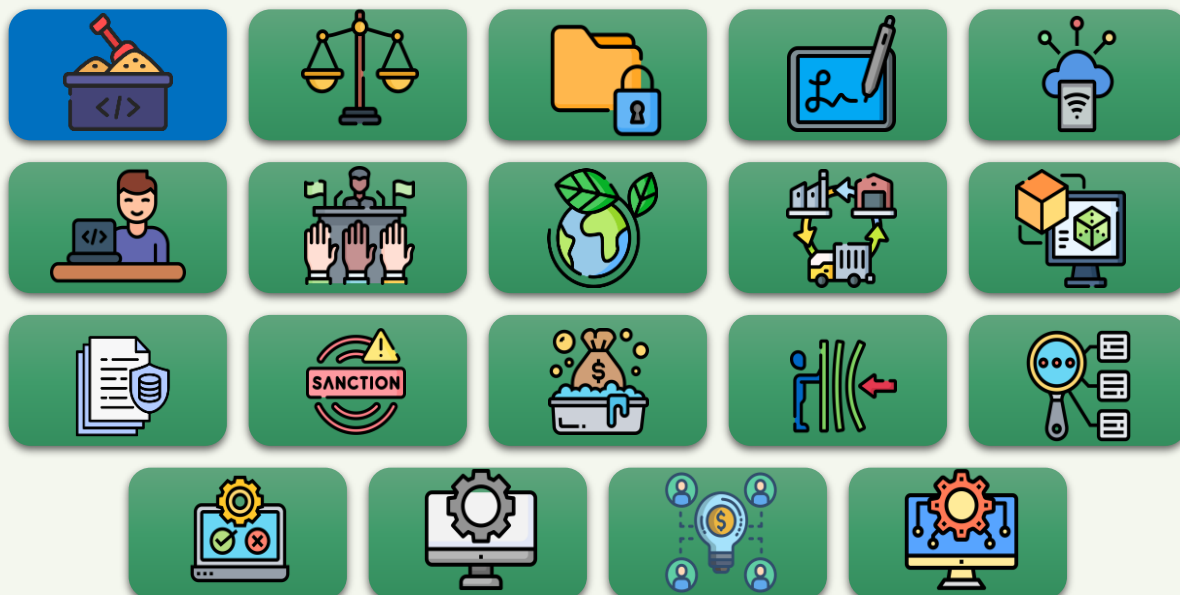


Understanding AI Regulatory Sandboxes





1. Introduction (Cohort Overview)



What lessons can be applied?

- AI Regulatory Sandboxes under the EU AI Act benefit from a structured, multi-phase engagement model.
- Lessons from similar frameworks show that early and iterative regulatory dialogue helps innovators address compliance issues, especially in cross-border AI deployments.
- Involving a wide range of competent authorities, from data protection to product safety, supports comprehensive feedback.
- Using sandbox protocols that include onboarding, regulator-only sessions, and tiered discussions fosters a safe, pre-commercial testing environment.
- Strong legal framing avoids uncertainty and promotes AI system design that aligns with fundamental rights, data minimization, transparency, and the high-risk provisions of the AI Act.

What are best practice examples?

- Best practices include: regulator-only meetings between AI sandbox sessions to allow reflection and coordination on emerging compliance questions; early identification of applicable provisions under the AI Act (e.g. Title III obligations); modular onboarding that tailors guidance to sector and risk class; and structured legal mapping to determine overlaps with GDPR, DSA, and sector-specific laws.
- Pilots combining AI with IoT, edge computing, or DLT benefitted from multi-regulator input.
- Pre-defined metrics for lawful AI development—such as data governance, transparency logs, and human oversight—enabled regulatory clarity and enhanced mutual understanding between developers and authorities.

What are relevant recommendations?

- Establishing AI sandboxes requires robust procedural design.
- Recommendations include setting clear eligibility criteria based on AI Act risk classification, providing tailored guidance on high-risk system requirements (Articles 9–15), & coordinating with national data protection authorities and market surveillance bodies.
- Encourage the use of harmonized technical standards where available, while allowing for case-specific flexibility in a sandbox context.
- Regulatory feedback should be documented and shared (in anonymized form) to contribute to consistent enforcement practices and policymaking across Member States.
- Aligning sandbox learnings with notified bodies and conformity assessment mechanisms will also strengthen post-sandbox compliance pathways.





2. Terminology, Laws, and Regulatory Framework



What lessons can be applied?

- AI Regulatory Sandboxes must establish a harmonized terminology aligned with the EU AI Act and related regulations.
- A key lesson from earlier sandboxes is that unclear or inconsistent use of legal and technical definitions (e.g., "AI system," "high-risk," "deployers") hampers effective regulatory dialogue.
- Mapping the full spectrum of applicable EU legislation—such as the AI Act, GDPR, eIDAS 2, and the DSA—is essential to identify compliance obligations.
- Recognizing overlaps (e.g., between data protection and AI training data governance) ensures that participants can address legal risks holistically from system design to deployment.

What are best practice examples?

- Effective sandboxes apply a layered regulatory framework early in the process.
- This includes clarifying whether the AI system is subject to Title II (prohibited practices), Title III (high-risk systems), or Title V (transparency obligations).
- Best practices include using shared glossaries from the AI Act and supporting guidance, and involving domain-specific regulators to interpret intersecting rules (e.g., health, finance).
- Coordinated legal mapping of AI lifecycle stages with applicable obligations (e.g., Article 10 for data governance, Article 14 for human oversight) enables structured compliance planning and meaningful, cross-sectoral regulatory feedback during sandbox operation.

What are relevant recommendations?

- Future AI sandboxes should adopt and require a shared terminology aligned with the AI Act's Article 3 definitions.
- Regulatory analysis should clearly delineate each system's lifecycle stage and its associated obligations (development, testing, deployment).
- Guidance should include mappings to overlapping regimes such as the GDPR, DSA, and sectoral safety laws (e.g., medical device regulation).
- AI sandboxes should provide early-stage legal risk assessments and identify standards or common specifications under Article 40.
- A formal documentation framework should track interpretative insights to support consistent application across Member States and feed into harmonized implementation.



3. Data Protection (GDPR)



What lessons can be applied?

- AI systems, especially high-risk ones under the AI Act, often process personal data during training or deployment. Sandbox dialogues show GDPR compliance must be built in early, via data minimization, role clarity, and lawful processing.
- Common challenges include anonymization vs. pseudonymization, federated learning roles, and Article 22 rights (**See Section 15**).
- For instance, a healthcare AI predicting patient outcomes must assess whether its data use triggers full GDPR obligations.
- Addressing these issues early ensures lawful development and avoids systemic compliance failures.

What are best practice examples?

- Best practices include requiring participants to perform detailed data mapping exercises and DPIAs aligned with Article 35 GDPR.
- Structured guidance was provided on lawful bases (e.g., public interest, contractual necessity) and the incompatibility of consent in certain AI contexts.
- Sandbox use cases involving biometric and behavioral data applied layered governance models, ensuring role identification (controllers vs. processors) and establishing contractual frameworks.
- Use of technical safeguards such as differential privacy and synthetic datasets was promoted to reduce reliance on identifiable data.

What are relevant recommendations?

- AI sandboxes should mandate early-stage GDPR risk assessments tailored to AI-specific processing, particularly for training data, real-time inference, and automated decision-making.
- Regulatory authorities should co-develop templates for role classification (controller, processor, joint-controller) and clarify expectations around data subject rights in AI contexts.
- Ensure interoperability between GDPR and AI Act requirements, especially Article 10 (data governance) and Article 29 (post-market monitoring).
- Regulators should offer guidance on managing legacy datasets, data minimization in model retraining, and anonymization techniques.



4. Relevance of eIDAS and Trust in AI Systems



What lessons can be applied?

- The integration of AI systems with digital identity frameworks (e.g., EUDI Wallet) and trust services under eIDAS 2 is becoming critical.
- Lessons from regulatory sandboxes show that trust in AI outputs often hinges on secure authentication, verifiable credentials, and auditability.
- AI systems interfacing with identity services (e.g. for access control or electronic signatures) must comply with eIDAS trust requirements.
- Early coordination between AI developers and trust service providers is essential.
- Misalignment between AI lifecycle needs and identity assurance protocols can delay deployment.

What are best practice examples?

- Best practices include using qualified trust services for verifying AI-generated documents or decisions (e.g. timestamping), and leveraging EUDI Wallets for identity verification in high-risk AI applications.
- AI use cases involving automated access or authorization decisions integrated eIDAS-based identity layers to enhance legal certainty.
- Sandbox participants explored the use of verifiable credentials and decentralized identifiers (DIDs) in alignment with Article 10 AI Act (data governance) and eIDAS 2.
- Coordination with national supervisory bodies helped clarify legal validity of AI outputs in human-in-the-loop.

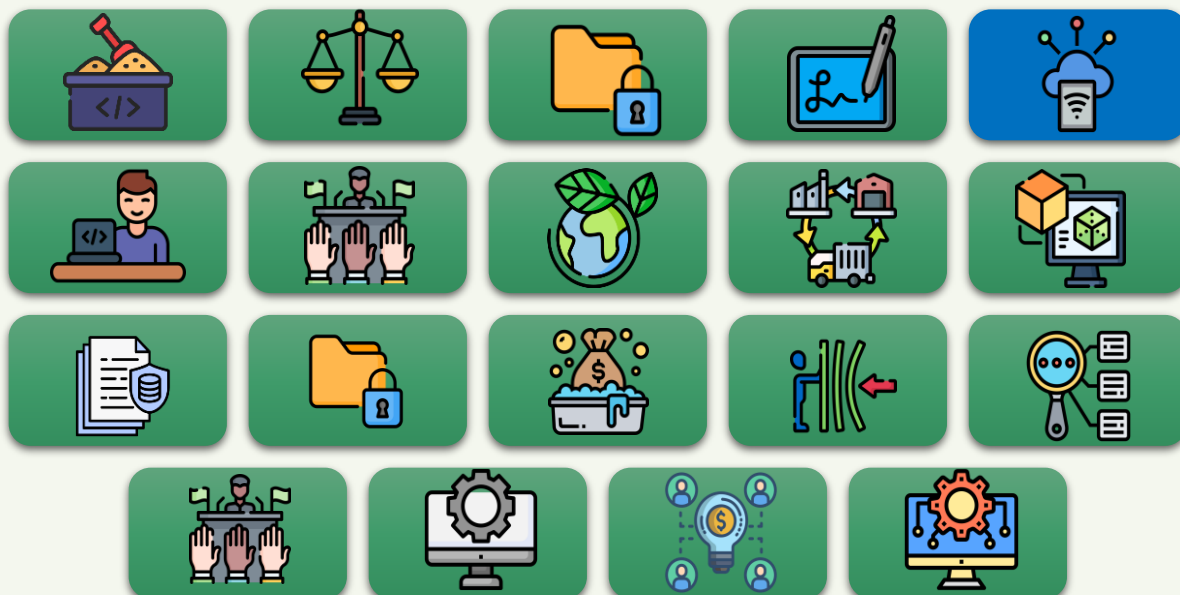
What are relevant recommendations?

- AI regulatory sandboxes should incorporate identity and trust layers early in system design, particularly for high-risk AI applications that rely on authentication or generate legally binding outputs.
- Participants should map eIDAS-relevant roles (e.g., trust service provider vs. AI deployer) and ensure interoperability between trust frameworks and AI logs or risk management systems.
- Regulators should provide guidance on integrating qualified electronic attestation of attributes (QAA) into AI-driven access or authorization systems.
- Future sandboxes should test EUDI Wallet integration for both AI user and system identity.





5. AI & Enabling Technologies (e.g. IoT)



What lessons can be applied?

- AI systems increasingly operate in conjunction with enabling technologies such as IoT, edge computing, and distributed ledger technologies (DLT). Sandboxes show that legal and compliance risks escalate with system complexity—especially regarding data provenance.
- When combined, these technologies introduce cross-cutting regulatory issues involving the AI Act, GDPR, & eIDAS 2.
- Lessons include the need for early regulatory mapping across the full technology stack and proactive identification of governance responsibilities, particularly for data flows across IoT devices or immutable DLT infrastructures feeding into AI decision-making systems.

What are best practice examples?

- Best practices involve establishing layered risk assessments that account for each component in the tech stack—e.g., smart sensors feeding AI algorithms that trigger automated decisions recorded on a blockchain.
- Sandboxes supported modular compliance strategies by matching relevant regulators across sectors (data protection, financial services, product safety).
- Participants used data governance frameworks (aligned with AI Act Article 10) to validate IoT data integrity and explored liability allocation in federated AI-DLT systems.

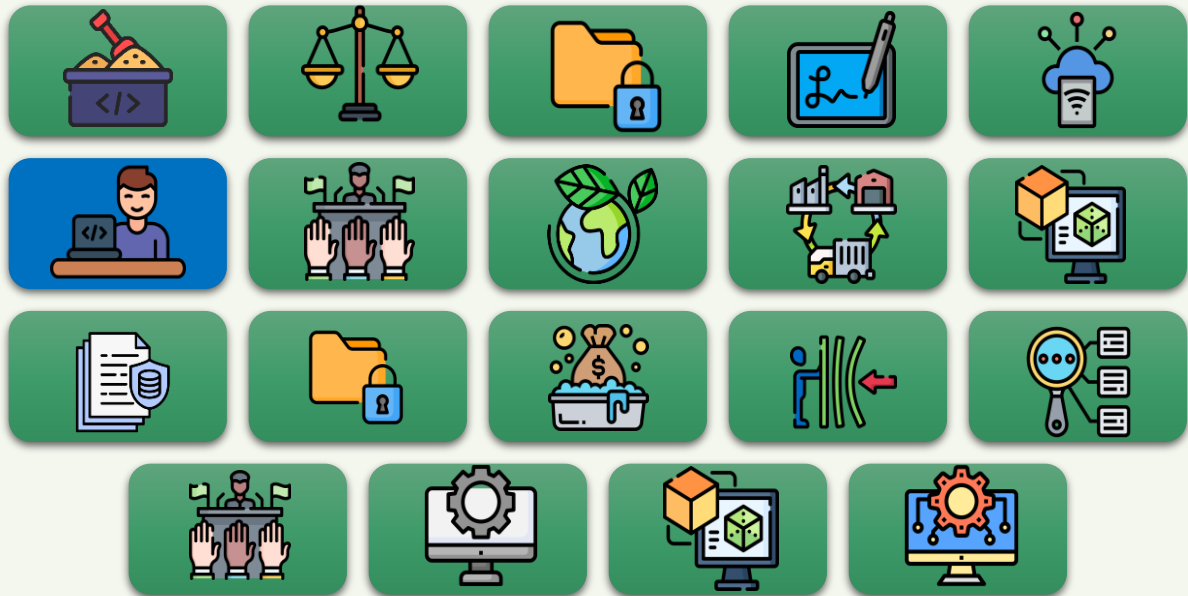
What are relevant recommendations?

- AI sandboxes should require a system-level view of AI deployments integrated with enabling technologies. Participants must identify and classify the risk and compliance posture of each component (AI, IoT, DLT) under applicable EU laws.
- Regulatory guidance should clarify shared responsibility models for AI systems built on decentralized or sensor-fed infrastructures.
- Promote use of harmonized standards for data integrity, traceability, and auditability across integrated systems.
- Sandboxes should include technical validators to assess how well AI governance mechanisms operate in complex, multi-technology environments.





6. Legal Entity Structures for AI Model Providers



What lessons can be applied?

- The AI Act applies legal obligations differently depending on whether an actor qualifies as a provider, deployer, importer, or distributor.
- Lessons from sandbox experience show that legal uncertainty around AI supply chains—especially in foundation models and open-source ecosystems—complicates role allocation and liability. Entity structure (e.g., consortium) significantly impacts compliance strategy, governance obligations, and market access.
- Sandboxes reveal the need for clear contractual frameworks and corporate structuring aligned with AI Act responsibilities, particularly in collaborative development settings or when foundation models are fine-tuned..

What are best practice examples?

- Best practices include sandbox participants formally identifying their legal role(s) under Article 3 of the AI Act and clarifying contractual accountability, especially for joint development or multi-actor foundation model pipelines.
- Where foundation models are released open-source, developers specified licensing models and risk disclaimers aligned with AI Act Title IX. Innovative entity structures such as consortia or cooperatives used governance frameworks that embedded AI compliance oversight, including post-market monitoring functions.
- Legal wrappers for decentralized or academic-led projects were evaluated to ensure assignable responsibility and traceability throughout the AI lifecycle.

What are relevant recommendations?

- AI regulatory sandboxes should support early legal role mapping for developers and deployers, particularly in foundation model & open-source contexts.
- Provide model contracts or templates that address liability, compliance delegation, and responsibilities under Articles 16–29 of the AI Act.
- Encourage legal structuring that allows for compliance continuity post-market (e.g., in case of model reuse or transfer). Clarify how Article 52 transparency obligations and risk management duties apply to foundation model developers vs. downstream implementers.
- Recommend sandbox collaboration with corporate law experts and authorities to align emerging AI business models with regulatory accountability.



7. AI in Democratic Processes



What lessons can be applied?

- AI applications in democratic processes—such as e-voting systems, civic engagement platforms, and deliberative democracy tools—raise high regulatory and ethical stakes.
- Lessons from sandbox dialogues emphasize the need to ensure transparency, accountability, and human oversight when AI is used to process or influence political participation.
- Use of AI in these contexts may constitute high-risk systems under Annex III of the AI Act, especially when impacting fundamental rights.
- Legal safeguards, including auditability, non-discrimination, and explainability, must be embedded by design.

What are best practice examples?

- Best practices include applying Article 14 (human oversight) and Article 13 (transparency) rigorously in systems that support or inform democratic participation.
- Projects used explainable AI techniques to ensure user trust and accountability in recommendation engines and moderation tools.
- Sandboxes facilitated collaboration with electoral commissions and digital identity providers to validate the secure use of authentication mechanisms, including potential eIDAS/EUDI Wallet integration.

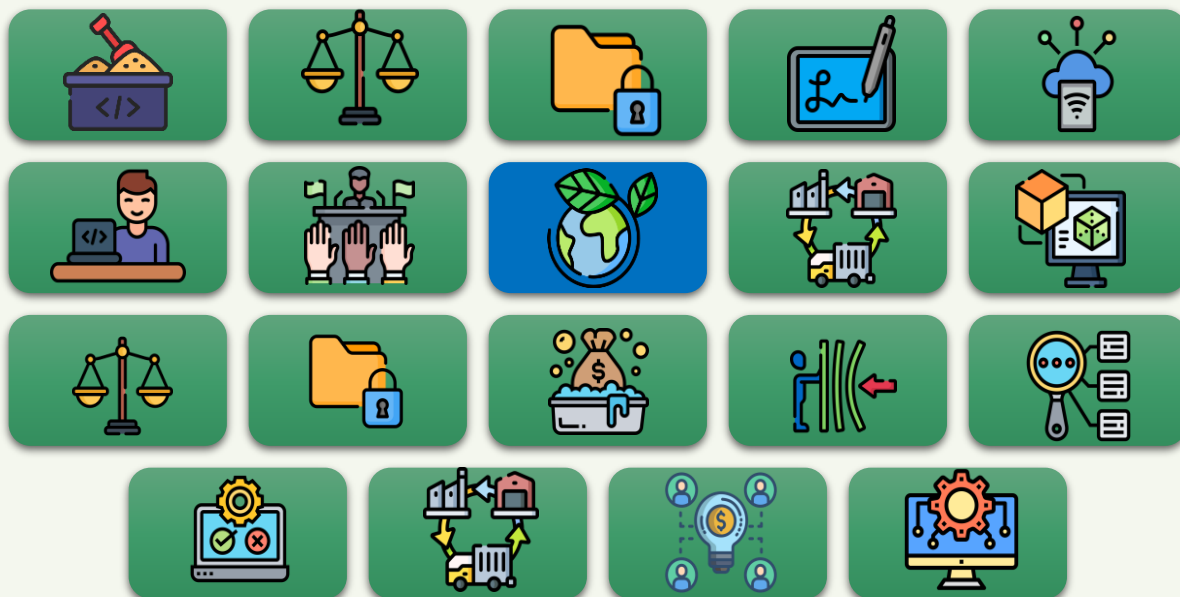
What are relevant recommendations?

- AI regulatory sandboxes should treat democratic use cases as a priority for scrutiny and cross-agency engagement.
- They should guide participants through risk classification under Annex III and require fundamental rights impact assessments in parallel with Article 29 obligations.
- Encourage development of sector-specific codes of conduct and redress mechanisms tailored to civic applications.
- Regulators should coordinate with electoral bodies, fundamental rights institutions, and cybersecurity agencies to ensure robustness, resilience, and public trust.





8. AI for ESG and Sustainability Reporting



What lessons can be applied?

- Lessons from sandbox dialogues show that early regulatory engagement helps clarify complex frameworks like the CSRD.
- Applied to AI Regulatory Sandboxes, iterative feedback from competent authorities ensures AI systems used for sustainability reporting align with legal requirements from the outset.
- Clarity on data provenance, auditability, and the verifiability of AI-driven outputs is essential.
- Developers learned that building regulatory trust requires transparent documentation and early integration of ESG data standards.

What are best practice examples?

- Best practices include co-developing AI models with regulators through structured sandbox dialogues, as seen in DLT-based CSRD reporting tools.
- A key example is the use of blockchain to ensure immutability of AI-generated ESG data, which improved trust among authorities.
- Translating this to AI sandboxes, firms should ensure traceability of model outputs and document ESG alignment using harmonized ESRS frameworks.
- Another effective practice is involving national competent authorities early to review data governance and explainability features, facilitating smoother scaling and approval.

What are relevant recommendations?

- AI developers should implement robust internal controls to validate that ESG metrics generated by AI systems conform to CSRD and ESRS standards.
- Regulatory Sandboxes under the AI Act should include CSRD-specific sessions with sustainability experts and regulators. It's recommended to align AI training data with standardized ESG taxonomies and integrate mechanisms to audit algorithmic decisions.
- Regulators should provide preemptive guidance on acceptable AI methodologies for CSRD use.
- Developers are also advised to anticipate cross-border regulatory variations and maintain flexibility for system updates to reflect evolving EU sustainability reporting norms.





9. AI in Supply Chain Due-Diligence



What lessons can be applied?

- Lessons from blockchain-enhanced EUDR compliance highlight that regulatory sandboxes offer a secure space to test due diligence mechanisms using innovative tech.
- For AI sandboxes, this means stress-testing AI systems for traceability, reliability, and legal alignment under real-world conditions.
- Early engagement with environmental and data protection regulators is vital, as is establishing transparency in AI-generated deforestation risk scores.
- Developers learned that explainability of supply chain assessments is critical for credibility and legal defensibility. Systems must also adapt to diverse data sources.

What are best practice examples?

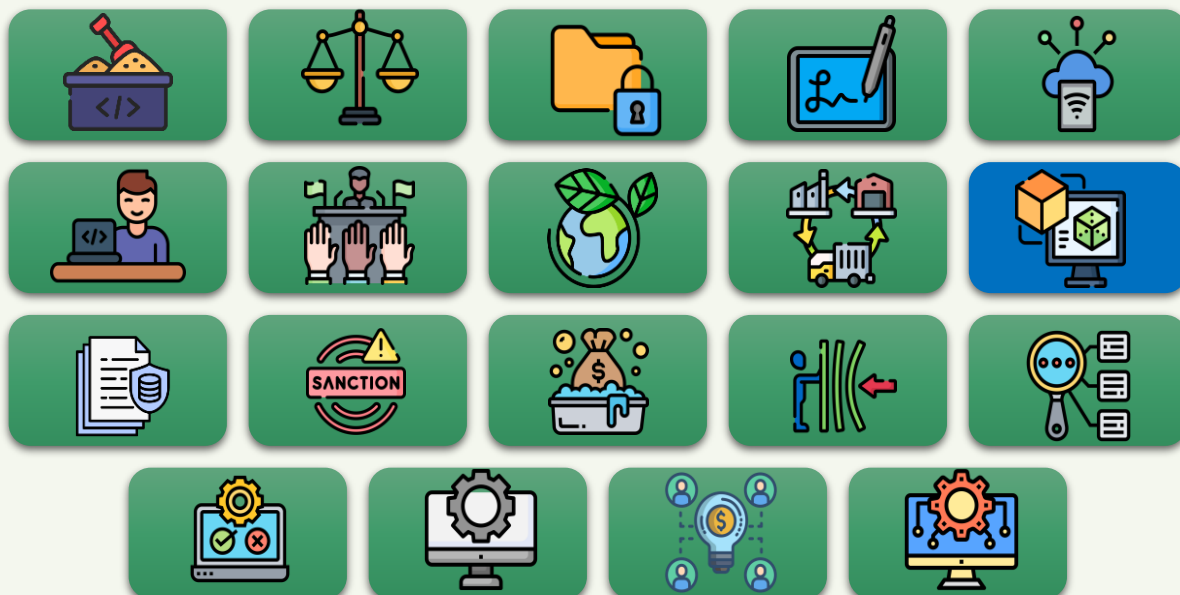
- In the blockchain context, best practices included using satellite data and geospatial tools to verify supply chains and log immutable records for timber and soy compliance.
- Applied to AI, sandboxes should support integration with verifiable datasets and standard due diligence metrics. Simulating end-to-end supply chain risk using synthetic data helps ensure transparency and auditability.
- Collaborating with environmental agencies and NGOs improved accuracy and regulatory alignment.
- Networks like the European Forum for Innovation Facilitators (EFIF) can help standardise sandbox practices and prevent market fragmentation across Member States.

What are relevant recommendations?

- AI systems used for EUDR compliance should incorporate explainable models capable of justifying deforestation risk flags with reference to geospatial and contextual evidence.
- Regulatory sandboxes should involve environmental law experts and Member State authorities to clarify national interpretations of due diligence thresholds.
- It's recommended to embed traceability features that align with EU deforestation data protocols and enable cross-border supply chain mapping.
- AI developers should also anticipate evolving EUDR requirements by designing flexible compliance modules and documenting model decisions in user-friendly formats.



10. Product Passports, Smart Labelling, and Digital Twins



What lessons can be applied?

- From sandbox use cases involving blockchain-enabled digital product passports (DPPs), a key lesson is the value of standardized data structures and persistent identifiers for interoperability.
- For AI systems, this translates into aligning output formats and metadata with ESPR and DPP requirements early in development.
- Regulatory sandboxes allow for testing AI-generated lifecycle data, simulations (digital twins), and labelling against evolving EU product sustainability standards.
- Lessons also show that regulators appreciate proactive measures to address traceability, version control, and long-term data accessibility—essential for AI systems that continually update data.

What are best practice examples?

- A best practice in blockchain DPP use cases was enabling continuous, verifiable updates of product data throughout the lifecycle using decentralized identifiers and versioned digital records.
- For AI systems, sandboxes should replicate this by testing lifecycle simulations (digital twins) and smart labels under real usage scenarios, ensuring conformity with the ESPR.
- Developers have successfully piloted AI models that predict degradation or compliance risks based on usage patterns, validated through regulatory dialogue.
- Early alignment with ESPR annexes and collaboration with industry consortia to harmonize taxonomy and data formats are recommended practices.

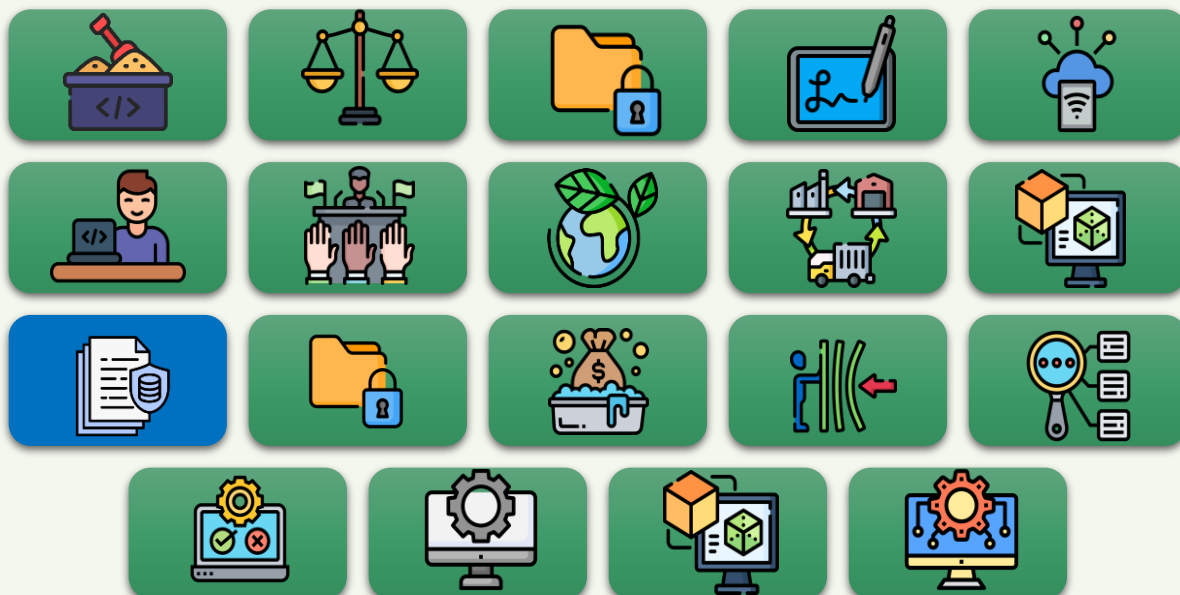
What are relevant recommendations?

- AI systems supporting DPPs should embed lifecycle traceability, including version-controlled data logs and mechanisms for auditability.
- Regulatory sandboxes should include experts in the ESPR, standardization bodies, and consumer product safety to ensure smart labelling and digital twin models align with EU sustainability and circular economy goals.
- It is recommended to test how AI-generated insights (e.g. predictive maintenance or recyclability scoring) influence regulatory labelling obligations. Developers should ensure AI outputs are explainable and machine-readable while also accessible to non-technical actors like regulators, consumers, and repair operators.





11. Digital Governance Act & Digital Services Act



What lessons can be applied?

- From blockchain sandbox dialogues, a core lesson is that early legal qualification of platform roles under the DGA and DSA is essential.
- Applied to AI sandboxes, this means clarifying whether AI-enabled platforms function as data intermediaries or online intermediaries subject to DSA obligations.
- Lessons show that integrating transparency, data-sharing governance, and accountability mechanisms early on prevents rework. Importantly, regulatory sandboxes allow AI developers to explore lawful data reuse scenarios under the DGA and test algorithmic transparency obligations under the DSA in a supervised setting, particularly where AI systems rank user-facing content.

What are best practice examples?

- A best practice from DLT-based sandboxes was using registries to categorize data holder roles under the DGA and disclose recommender system logic under the DSA.
- AI sandboxes can mirror this by testing data-sharing protocols with public sector bodies and ensuring AI-generated content complies with DSA Articles 27–30.
- Another effective practice is performing risk-tier classification for AI systems embedded in intermediary services. Early collaboration with national coordinators and data altruism bodies helped clarify edge cases.

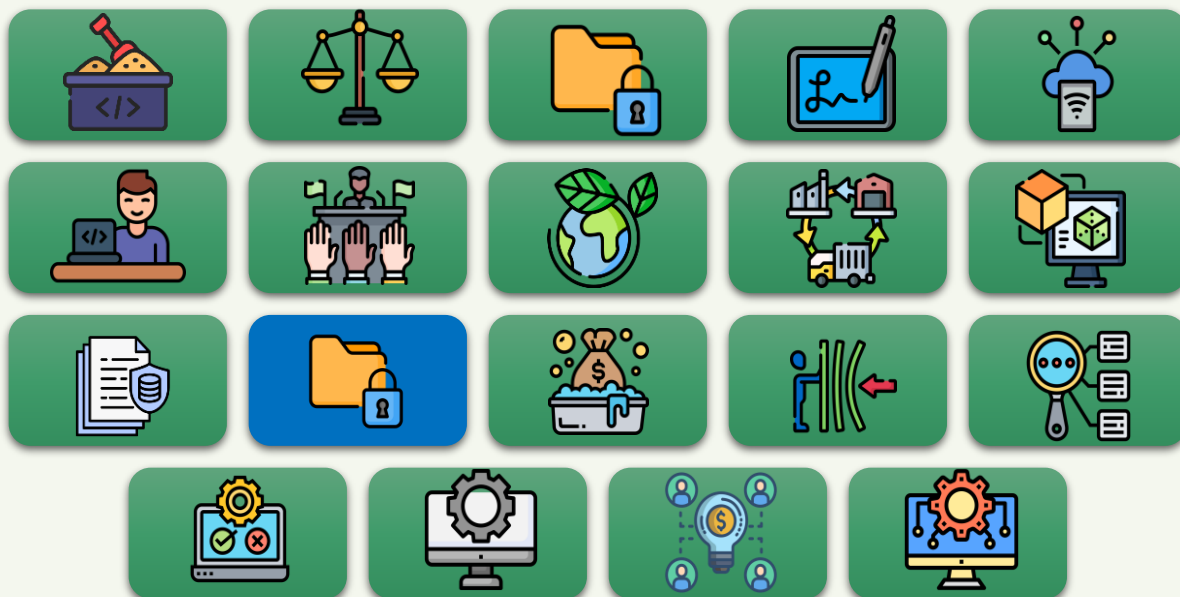
What are relevant recommendations?

- AI developers should assess if their systems qualify as Very Large Online Platforms (VLOPs) or intermediary services under the DSA and plan for systemic risk mitigation, transparency reporting, and algorithmic auditing accordingly.
- In the DGA context, those facilitating data reuse must clarify legal roles and comply with interoperability and user consent requirements.
- Regulatory sandboxes should involve DGA/DSA competent authorities to pre-validate governance models, data-sharing conditions, and AI explainability practices.
- It is recommended to develop machine-readable policy disclosures and real-time explainability features for AI.





12. AI in Sanctions Screening & Compliance Automation



What lessons can be applied?

- Lessons from blockchain-enabled sanctions compliance emphasize the critical need for auditability, up-to-date sanctions list integration, and transparent logic in screening systems.
- Applied to AI sandboxes, developers learned that explainable flagging, traceability of inputs, and real-time adaptability are key to regulatory acceptance.
- Testing AI systems in sandbox environments helped identify misalignment with EU or UN sanctions frameworks and enabled refinement of false positive handling.
- Regulatory authorities stressed that sanction-related AI tools must balance efficiency with due process.

What are best practice examples?

- Best practices include embedding dynamic linkages to official sanctions databases and version-controlled audit logs of AI-generated alerts.
- In sandbox settings, AI developers simulated real-world compliance workflows and validated alert accuracy and escalation logic with competent authorities.
- High-performing projects applied tiered risk scoring to reduce false positives and used hybrid human-AI review loops to meet proportionality and accountability standards.
- Another key practice was including multilingual name matching and transliteration modules, which improved precision across jurisdictions.

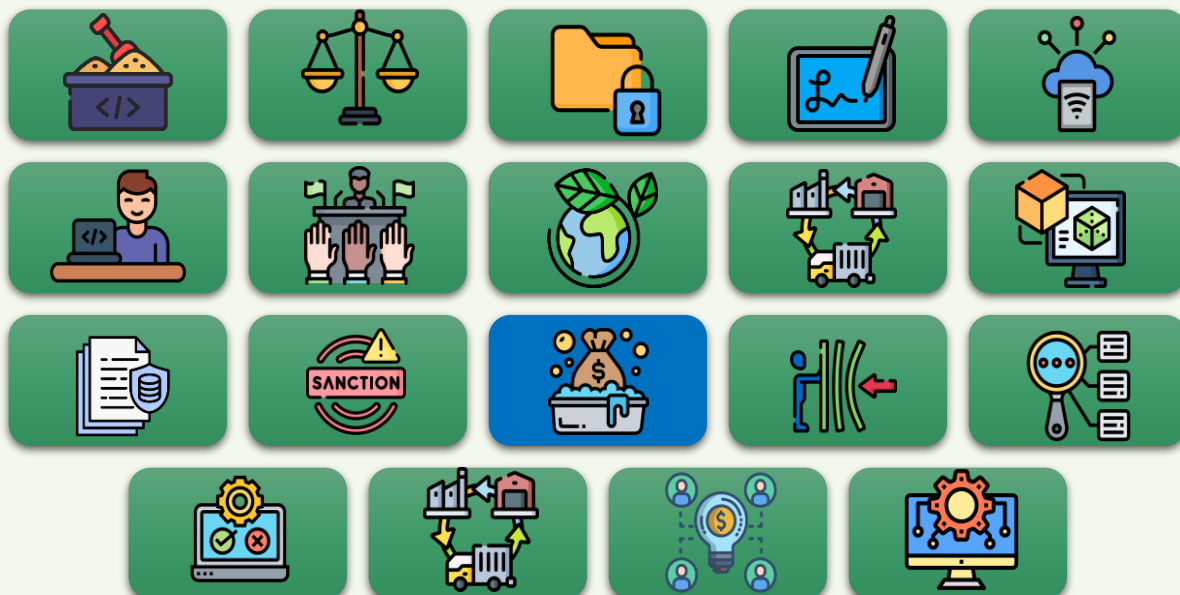
What are relevant recommendations?

- AI systems for sanctions screening should implement explainable models with traceable reasoning paths and standardized audit logs.
- It is recommended that AI developers test these systems under real-world scenarios using historical data in a sandbox setting, with oversight from relevant sanctions authorities. Integration with official EU and UN sanctions list APIs, periodic model updates, and risk-based alert thresholds should be core compliance features.
- Regulatory sandboxes should prioritize dialogues on edge-case interpretation, transparency obligations, and cross-border enforcement implications.





13. AI and Anti-Money Laundering (AML)



What lessons can be applied?

- Sandbox dialogues show that AI systems used in AML must balance innovation with strict regulatory expectations on accountability and transparency.
- A major lesson is that black-box AI models—especially in transaction monitoring—face skepticism without clear explanations of risk classification. Within regulatory sandboxes, developers learned to simulate high-risk scenarios and validate AI outputs against supervisory benchmarks.
- Cross-border AML rules require adaptable compliance logic, and early collaboration with national Financial Intelligence Units (FIUs) proved critical.

What are best practice examples?

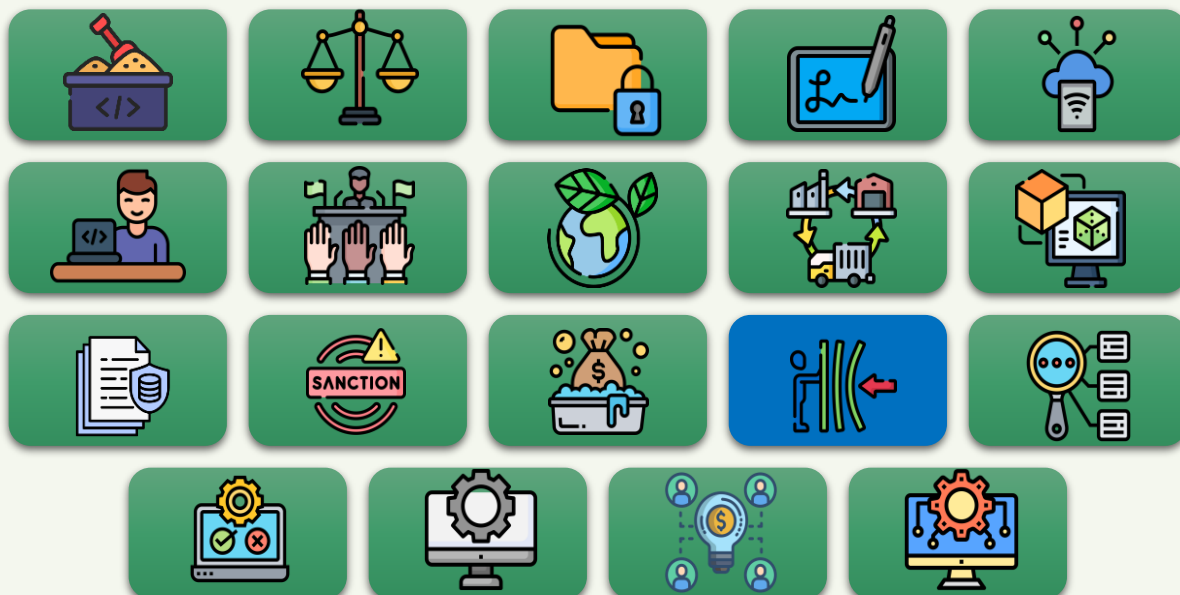
- Best practices include deploying layered AI models with rule-based preprocessing to ensure baseline compliance before applying advanced detection algorithms.
- Successful sandbox participants involved AML authorities to test explainability, flagging mechanisms, and risk scoring logic. Integrating transaction context, customer profiles, and network behavior enhanced suspicious activity detection while reducing false positives.
- Continuous model validation, supported by audit logs and human-in-the-loop review, enabled better regulatory confidence.

What are relevant recommendations?

- AI developers should implement transparent risk-scoring models that allow compliance officers to understand and explain why a transaction was flagged.
- Regulatory sandboxes should facilitate collaboration with FIUs, AML supervisors, and data protection authorities to clarify roles, responsibilities, and data-sharing thresholds.
- It's recommended to embed features for continuous model monitoring, document rationale for flagged transactions, and ensure explainability for both regulators and auditors.
- Developers should anticipate dual compliance with the AML Regulation and AI Act by designing human oversight workflows accessible for investigations



14. Digital Operational Resilience (DORA) for AI



What lessons can be applied?

- From sandbox discussions under DORA, a key lesson is that AI systems used in financial services must be treated as critical ICT components if they support important business functions.
- Regulatory sandboxes allow firms to assess how AI aligns with DORA's requirements for ICT risk management, incident response, and third-party oversight.
- Developers learned that operational resilience isn't just about uptime—it's about explainability, traceability, and maintaining control during disruptions.
- Testing in sandbox environments revealed that AI system failures must be monitored, logged, and remediated with governance mechanisms that mirror those for traditional ICT providers.

What are best practice examples?

- Best practices include mapping AI dependencies within financial service workflows to assess their criticality under DORA.
- Firms in sandboxes successfully simulated incident scenarios involving AI model drift or cyberattack-induced bias, and documented response protocols in line with DORA Articles 9–12.
- Another effective approach was integrating AI lifecycle management with enterprise ICT frameworks, ensuring alignment between model governance and operational resilience testing.
- Participants also piloted third-party risk assessments for outsourced AI components and clarified service-level agreements to meet DORA's obligations.

What are relevant recommendations?

- AI systems supporting critical financial functions should be included in Digital Operational Resilience strategies, with risk assessments, backup plans, and incident response workflows tailored for AI-specific failures.
- Regulatory sandboxes should involve financial supervisors and ICT risk experts to test DORA compliance in AI contexts.
- Developers are advised to implement real-time monitoring of AI performance, document model governance procedures, and conduct Threat-Led Penetration Testing (TLPT).
- Clear accountability and explainability frameworks should be embedded to ensure that disruptions can be contained and reported as per DORA's notification obligations.



15. Classification of AI Systems under EU AI Act



What lessons can be applied?

- A key lesson from sandbox testing is that correct classification of AI systems under the AI Act (e.g. high-risk, limited-risk, or prohibited) is foundational to compliance.
- Many developers underestimated the complexity of determining whether a system falls within Annex III (high-risk use cases), particularly in finance, employment, and biometric systems.
- For example, an AI tool used to rank job applicants raised questions about its classification and triggered stricter obligations.
- Sandboxes helped clarify such borderline cases and informed internal frameworks, legal reviews, and downstream duties, e.g. documentation.

What are best practice examples?

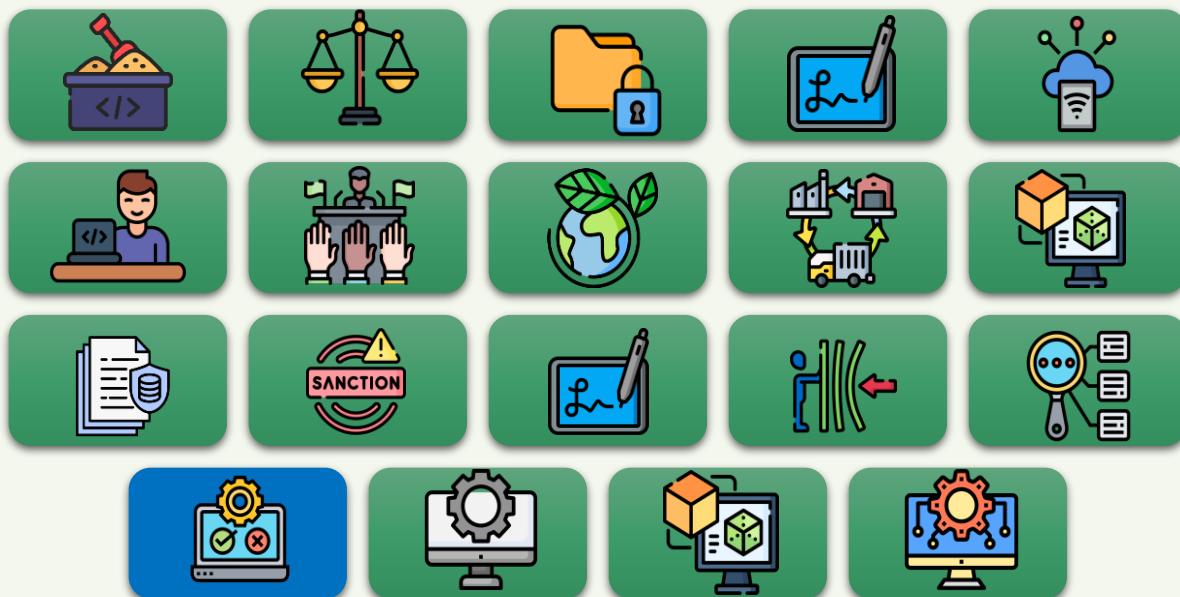
- Best practices include conducting detailed use-case mapping against Annex III categories and maintaining internal classification logs with justifications.
- In regulatory sandboxes, developers shared mock conformity assessments with authorities to confirm high-risk status and test technical documentation requirements.
- Use of impact assessments (like AI risk matrices or conformity checklists) proved valuable, especially when AI systems had multiple applications.
- Dialogue with regulators on function creep—when systems shift into high-risk territory—helped refine product design.

What are relevant recommendations?

- AI developers should conduct early, structured assessments of whether their systems meet the criteria of high-risk AI under the AI Act.
- Regulatory sandboxes should offer model classification workshops with competent authorities to validate interpretations of Annex III.
- It is recommended to maintain a classification logbook, include legal reviews, and document the logic used for exclusions or boundary decisions.
- For borderline systems, firms should prepare technical documentation and risk management plans as if the system were high-risk.



16. Real-World Testing and Sandbox Exemptions



What lessons can be applied?

- Sandbox dialogues confirmed that Article 57–59 exemptions for real-world testing are critical for validating high-risk AI systems under supervised, lawful conditions.
- A key lesson is that early coordination with competent authorities enables safe experimentation without breaching user rights or product safety laws.
- Firms learned that testing plans must be specific, risk-mitigated, and include fallback mechanisms.
- Sandboxes also helped clarify national implementation variations and practical conditions for exemptions.
- Developers should treat real-world testing not just as validation but as a core phase for identifying risks.

What are best practice examples?

- Best practices include drafting detailed test plans that define scope, participants, metrics, and emergency exit procedures, then pre-submitting them for authority review.
- Successful sandbox participants engaged in iterative dialogue with regulators to adjust testing parameters in line with AI Act Article 58 safeguards.
- Firms also used pseudonymized datasets and controlled environments to simulate real-world use without exposing end users to unnecessary risk.
- Another strong approach involved documenting test results using AI Act-compliant formats, including impact logs and feedback loops, to refine the system before full market entry.

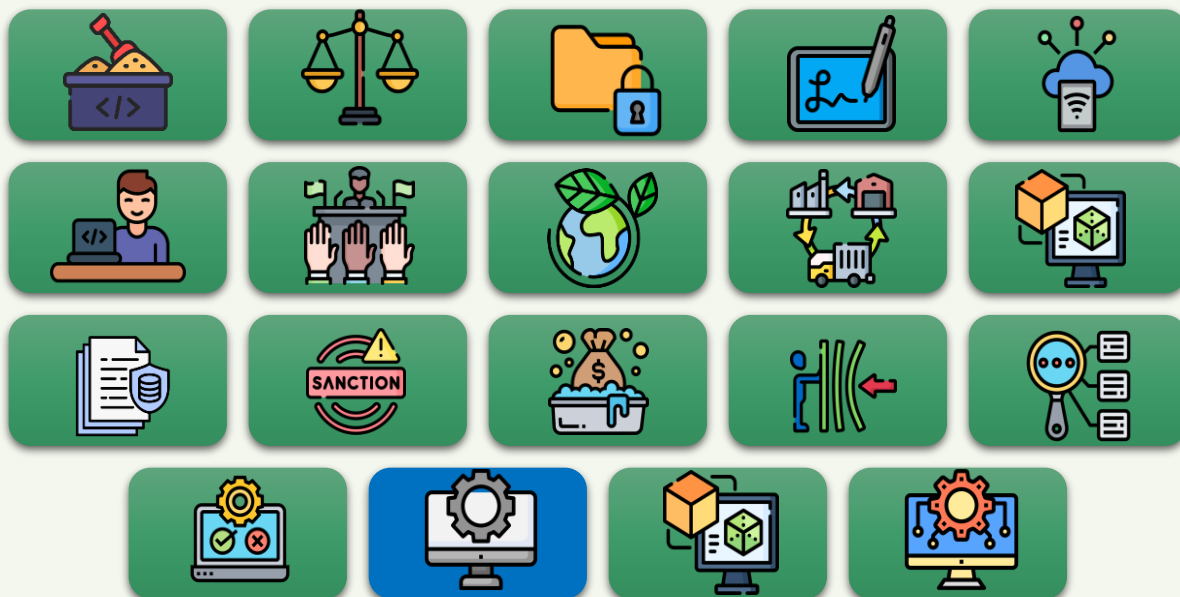
What are relevant recommendations?

- AI developers seeking Article 57–59 exemptions should prepare structured testing protocols with clear objectives, lawful data use, and measurable performance indicators.
- Regulatory sandboxes should facilitate early engagement with market surveillance authorities to pre-validate testing safeguards and compliance with fundamental rights.
- It is recommended to include consent mechanisms, risk mitigation strategies, and a plan for reporting adverse outcomes.
- Developers should also treat sandbox exemptions as an opportunity to test compliance documentation, transparency obligations, and human oversight frameworks.





17. Open Source AI – Roles & Responsibilities



What lessons can be applied?

- AI Regulatory Sandboxes revealed that roles and responsibilities around open-source AI are often unclear, particularly concerning liability and compliance in high-risk contexts.
- The AI Act distinguishes between free, non-monetized open-source AI and those used commercially.
- Lessons include the need for clarity on when open-source exemptions apply and how responsibility is shared among providers and third-party contributors.
- Effective sandbox participation helped identify these grey areas early, enabling iterative clarification and safe deployment of open models.

What are best practice examples?

- Best practices include using model cards and documentation (e.g., data sheets) to clarify the provenance and functionality of open-source AI components.
- Participants in the EU sandbox have begun adopting structured licensing and information-sharing protocols aligned with Article 25(4), ensuring that obligations on providers of high-risk AI systems are understood when integrating third-party open-source tools.
- Engaging legal and technical teams in sandbox iterations also helped navigate exemptions under Recital 103 without assuming liability exposure.

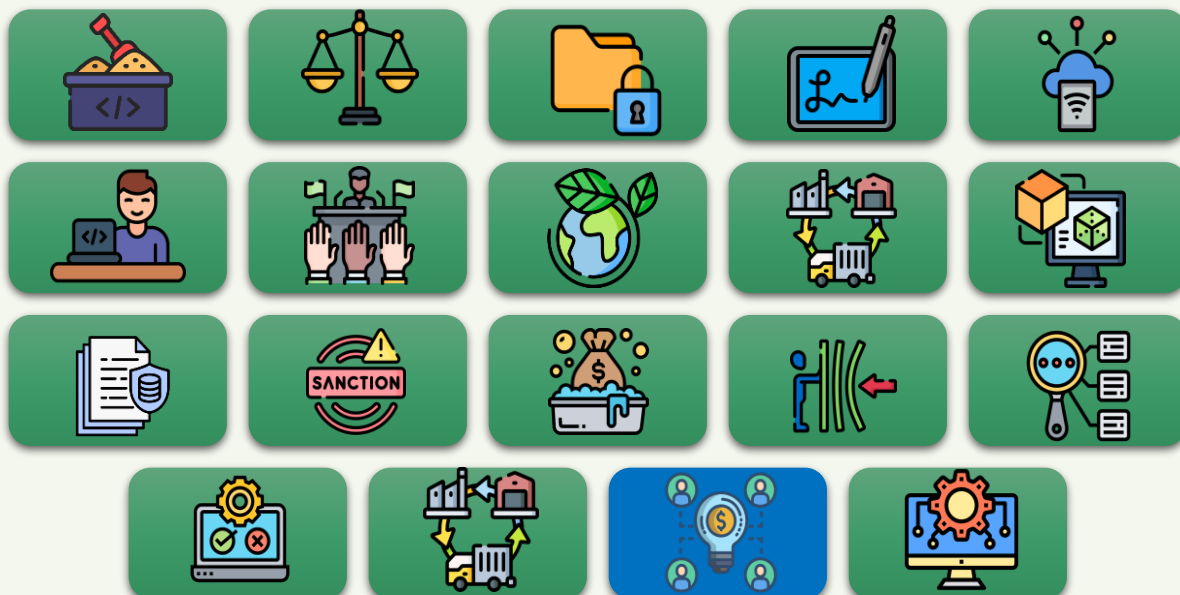
What are relevant recommendations?

- Regulatory sandboxes should encourage open-source AI developers to adopt standardized documentation practices and contractual frameworks, even where formal obligations may not apply.
- It's also recommended to clarify in advance whether integration of open-source components qualifies under the exemptions in Article 25(4) and to delineate responsibilities via clear agreements.
- Finally, national sandboxes should offer templates and decision trees for assessing applicability of open-source clauses under the AI Act to support SME participation and legal certainty.





18. AI for Crowdfunding, Finance, and Tokenised Assets



What lessons can be applied?

- AI Regulatory Sandboxes highlighted the complex interplay between the ECSPR, MiFID II, and MiCAR for AI-integrated crowdfunding platforms.
- Lessons include the need to clearly delineate custody responsibilities, capital requirements, and thresholds when AI supports tokenized assets or investor decision-making.
- Some sandboxes noted confusion over when a service is considered within ECSPR versus MiFID II scope, particularly with virtual asset integration.
- Sandbox iterations revealed that national divergences in custodial and accounting rules remain a challenge for pan-European platforms using AI for asset management.

What are best practice examples?

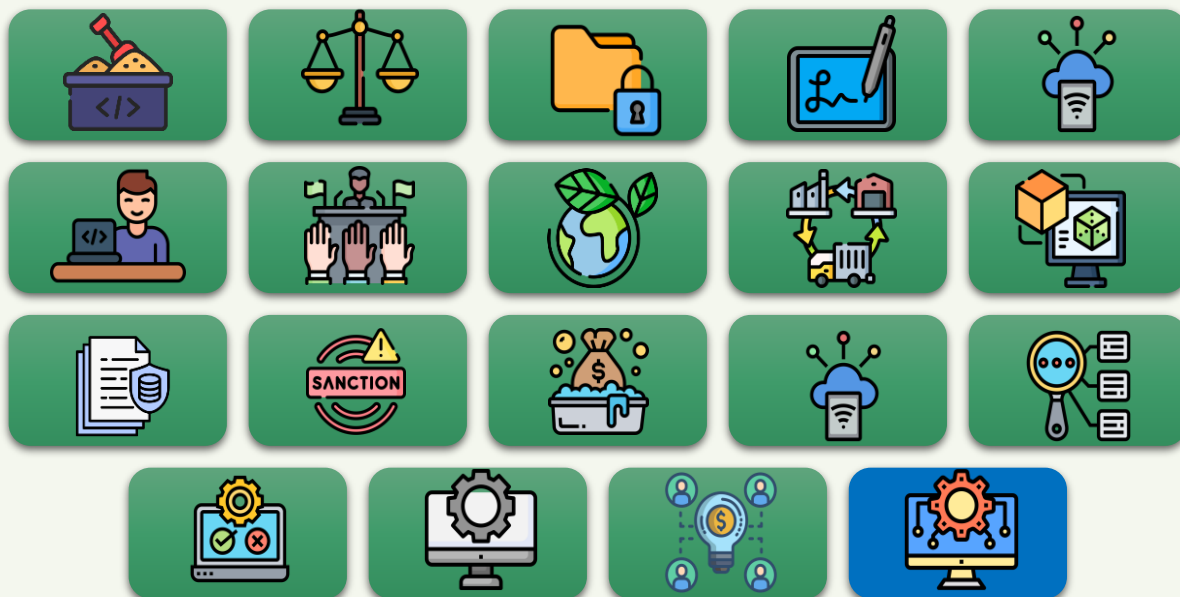
- Best practices include segregating AI functionalities from regulated services to avoid accidental triggering of custodial or licensing obligations.
- For example, firms integrated AI for investor profiling and risk scoring without letting the AI execute financial transactions, thus staying within ECSPR scope.
- Others used DLT sandboxes to pilot AI-based secondary trading ecosystems while remaining compliant with ECSPR and national safekeeping standards.
- Documenting AI-driven processes, ensuring transparency for retail investors, and engaging supervisors early during sandbox participation proved essential to ensure legal alignment.

What are relevant recommendations?

- It is recommended that firms developing AI for crowdfunding and tokenized asset platforms work closely with sandbox authorities to map out the regulatory perimeter—especially regarding MiFID II and MiCAR crossover.
- They should also adopt modular system designs where AI functionalities can be isolated if needed.
- Clear communication on safekeeping responsibilities under Article 10 ECSPR is vital.
- Sandboxes should issue guidance on hybrid business models and offer accounting templates for AI-managed virtual assets to reduce uncertainty around prudential capital treatment.



19. Automated Decision-Making & Algorithmic Markets



What lessons can be applied?

- The regulatory sandbox identified critical distinctions between AI-driven Automated Market Makers (AMMs) and traditional financial instruments under MiFID II.
- Lessons include recognizing when AMMs fall under the DLT Pilot Regime versus requiring a MiFID license. Importantly, integrating AI into algorithmic decision-making systems for financial markets often triggered high-risk classification under the AI Act, necessitating rigorous documentation, risk management, and explainability.
- Sandbox testing emphasized the need to clarify roles where AI executes autonomous financial actions with legal consequences—especially in environments lacking human oversight.

What are best practice examples?

- Best practices include segregating the AI decision layer from the execution layer to ensure human oversight and mitigate unintended market manipulation.
- One sandbox participant developed an AI-based risk model for liquidity management in AMMs, while keeping trade execution under supervisory control.
- Others ensured full compliance with MiFIR's transaction reporting and pre/post-trade transparency by integrating explainability tools that provided real-time rationales for AI-based decisions.

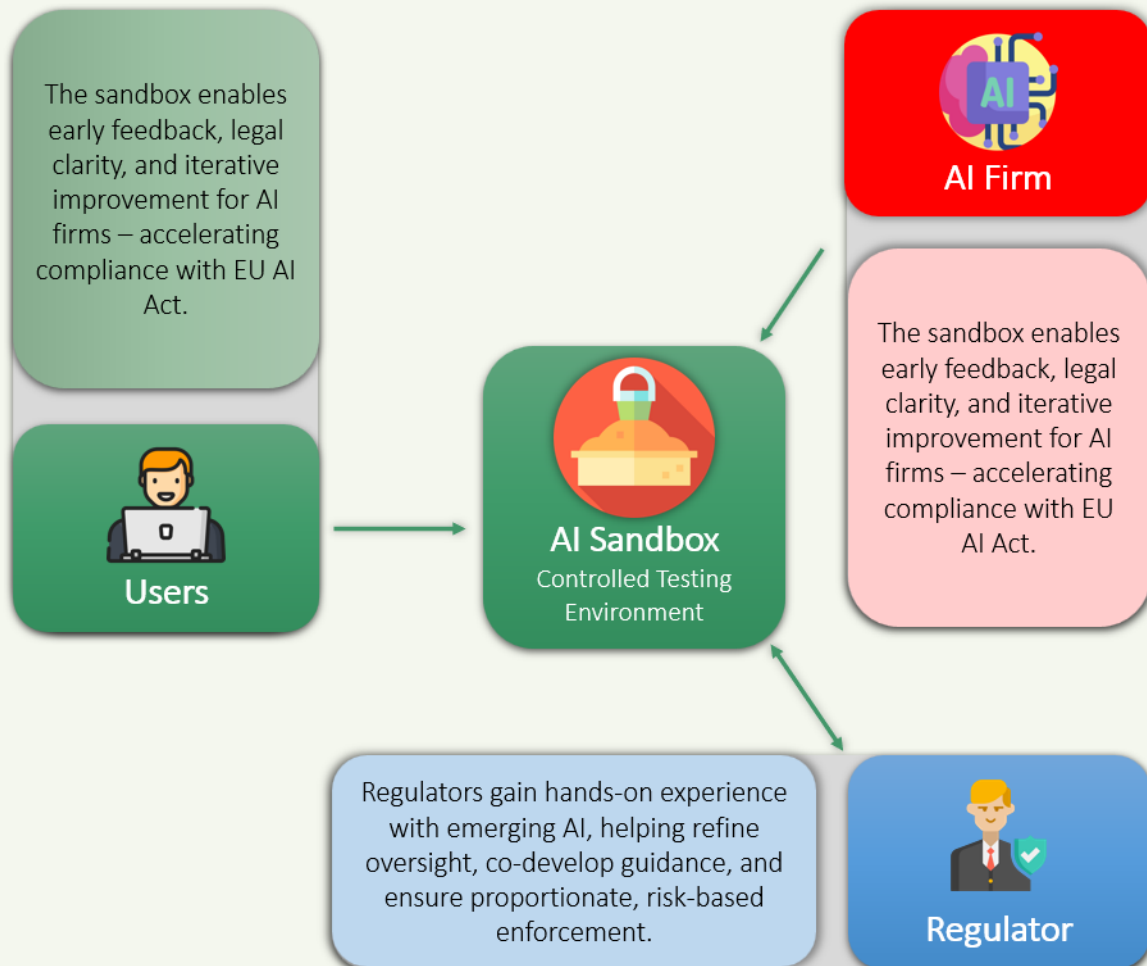
What are relevant recommendations?

- Firms developing AI for algorithmic markets should engage with regulators early to assess if their use case triggers obligations under MiFID II, MiCAR, or the DLT Pilot Regime.
- Regulatory sandboxes should encourage firms to prototype human-in-the-loop (HITL) mechanisms and offer guidance on reconciling AI system transparency with financial market fairness obligations.
- ESMA and national authorities are encouraged to publish interpretive guidance on AI's role in AMM structures and whether AI-based decision-making necessitates reclassification as an MTF or regulated market actor.



Mapping AI Regulatory Sandboxes to Ecosystem Actors







AI Regulatory Sandboxed		Actor		
Section	Description	AI Firm	Regulator	Users
1. Introduction (Cohort Overview)	Outlines the sandbox's purpose to facilitate safe regulatory experimentation for AI developers and regulators under the AI Act. Covers structure, participation, and dissemination of results.	The sandbox offers AI firms a structured, low-risk environment to test high-risk or innovative systems under regulatory supervision. It enables early feedback, legal clarity, and iterative improvement, accelerating compliance readiness and responsible innovation aligned with the EU AI Act.	Regulators gain hands-on exposure to emerging AI technologies, fostering practical understanding of compliance challenges. The sandbox enables them to refine supervisory practices, co-develop guidance, and build trust with innovators—ensuring proportionate, risk-based enforcement under the AI Act.	Users benefit indirectly as the sandbox promotes safer, more transparent AI systems. By ensuring early regulatory alignment, it reduces risks of harm, bias, or misuse in AI applications, enhancing user trust and accountability before systems reach the broader market.
2. Terminology, Relevant Laws, and Regulatory Framework	Defines key AI terms and maps out relevant legislation (AI Act, GDPR, DSA, etc.) affecting AI system development and deployment.	Understanding the legal definitions and regulatory scope—such as “high-risk,” “general-purpose,” or “provider”—is essential for AI firms. This section helps them classify systems correctly, navigate overlapping regulations (e.g. AI Act, GDPR, DSA), and prepare robust compliance strategies from development through deployment.	This section supports regulators in applying consistent interpretations across jurisdictions. It equips them to identify applicable laws, understand evolving AI terminology, and ensure proportional oversight. It also aids coordination when multiple frameworks (e.g. GDPR and AI Act) intersect in complex use cases.	Clear terminology and legal framing empower users to understand how AI systems that affect them are regulated. This promotes transparency around AI risk categories, user rights, and legal safeguards—enabling more informed engagement with AI technologies and better access to redress when needed.
3. Data Protection (GDPR) – Compliance in AI Contexts	Addresses GDPR compliance for AI, including lawful bases, explainability, data minimisation, and rights of data subjects.	Must ensure lawful data processing, uphold data minimisation, and provide explainability. Consent, legitimate interest, or contractual necessity must be	Oversees AI firm compliance, assessing risk, consent validity, and fairness. Provides guidance on GDPR application in complex AI scenarios, especially around	Gain rights to access, correct, or erase personal data processed by AI. Must be informed of data use, profiling logic, and receive meaningful explanations of AI-





		clearly justified, with robust mechanisms for data subject rights and transparency.	automated decisions and profiling.	driven decisions affecting them.
4. Relevance of eIDAS and Trust in AI Systems	Explores the role of eIDAS trust services in AI authentication, provenance, and digital signing, particularly for public and high-risk AI.	Must navigate overlapping regulatory requirements when integrating AI with technologies like IoT or DLT. Responsible for clarifying accountability, managing compound risks, and ensuring secure, compliant data flows. Should integrate eIDAS-compliant trust services (e.g., electronic signatures, seals) to ensure identity verification, data integrity, and secure transactions, particularly in high-risk AI systems requiring traceability and trust.	Ensures AI systems using trust services align with eIDAS standards. Assesses authentication mechanisms and their suitability for public services or critical applications involving verified digital identities.	Benefit from enhanced trust, transparency, and security when interacting with AI systems. eIDAS-backed services help ensure users' data is authenticated and actions are verifiable, reducing fraud and misuse risks.
5. AI + Enabling Technologies (e.g. IoT, DLT)	Discusses regulatory complexity when AI is combined with technologies like DLT or IoT, including shared accountability and compound risk.	Must navigate overlapping regulatory requirements when integrating AI with technologies like IoT or DLT. Responsible for clarifying accountability, managing compound risks, and ensuring secure, compliant data flows.	Assesses how combined technologies impact risk, liability, and compliance. Provides guidance on interpreting obligations across frameworks when AI is embedded in complex, multi-layered systems.	Face increased complexity and risk when using AI with IoT or DLT. Must be informed of data usage, security practices, and who is accountable in multi-party, tech-integrated environments.





6. Legal Entity Structures for AI Developers & Foundation Models	Reviews liability structures and legal wrappers (e.g. foundation model developers, open-source groups) for AI compliance under the AI Act.	Must choose appropriate legal structures to allocate liability, especially for foundation models and open-source contributions. Responsible for ensuring transparency, governance, and compliance under the AI Act.	Evaluates the legal form and accountability mechanisms of AI developers. Ensures clear responsibility for AI system outcomes, particularly when multiple actors or open-source models are involved.	Benefit from clarified accountability and legal recourse when interacting with AI systems. Understanding the developer's legal structure helps assess reliability, compliance, and who is liable in case of harm.
7. AI in Democratic Processes (e.g. e-Voting, Civic Tech)	Looks at AI in elections and civic engagement, focusing on transparency, fairness, and safeguards against bias.	Must design systems that uphold democratic principles, ensuring transparency, fairness, and resistance to manipulation or bias in civic applications like e-voting or public engagement platforms.	Monitors AI use in democratic contexts to prevent undue influence, misinformation, or bias. Sets standards for transparency, accountability, and auditability in AI-powered civic technologies.	Rely on trustworthy, fair AI systems to participate in democratic processes. Should be informed about how AI influences outcomes and have access to redress mechanisms in case of unfair treatment.
8. AI for ESG and Sustainability Reporting (CSRD Alignment)	Evaluates AI's use in ESG and sustainability reporting, especially for traceability, audit readiness, and data integrity.	Must ensure AI tools used in ESG reporting support data accuracy, traceability, and auditability, aligning with CSRD requirements and enabling reliable sustainability disclosures.	Assesses whether AI systems used in ESG contexts meet CSRD standards. Ensures transparency, verifiability, and integrity of sustainability data generated or processed by AI.	Depend on accurate, explainable AI-generated ESG data for informed decision-making. Should understand how AI influences sustainability metrics and trust that reporting complies with regulatory expectations.
9. AI in Supply Chain Due Diligence (e.g. EUDR)	Highlights AI use in monitoring environmental risks, deforestation, and supply chain transparency under the EUDR.	Must develop AI tools that monitor supply chains for risks like deforestation or human rights violations, ensuring traceability and compliance with regulations like the EUDR.	Verifies that AI systems used for due diligence provide reliable, transparent risk assessments. Ensures compliance with environmental and human rights obligations across global supply chains.	Rely on AI-driven insights to make ethical purchasing and sourcing decisions. Should be informed about how AI identifies risks and ensures supply chain transparency.





10. Product Passports, Smart Labelling & Digital Twins	Covers AI's role in product lifecycle traceability, labelling, and data quality under the Ecodesign and Battery Regulations.	Must design AI systems that support accurate product data, lifecycle tracking, and smart labelling in line with Ecodesign and Battery Regulations, enabling digital twins and product passports.	Ensures AI-enabled labelling and traceability tools meet regulatory standards for data quality, interoperability, and consumer transparency throughout the product lifecycle.	Benefit from clear, accessible product information enabled by AI. Can make informed choices based on environmental impact, durability, and origin, supported by digital twins and smart labels.
11. Data Governance Act & Digital Services Act	Considers if AI systems act as data intermediaries (DGA) or online platforms (DSA) and how to comply with related obligations.	Must determine whether their systems function as data intermediaries (DGA) or online platforms (DSA), and ensure compliance with relevant obligations around data sharing, transparency, and user rights.	Assesses the role of AI systems under DGA and DSA. Enforces rules on data access, content moderation, algorithmic transparency, and systemic risk mitigation where applicable.	Gain greater control over their data and platform interactions. Should be informed about how AI systems manage their data and content exposure, with channels for transparency and redress.
12. AI in Sanctions Screening & Compliance Automation	Examines AI in sensitive contexts such as sanctions and fraud screening, with a focus on explainability and proportionality.	Must design AI systems that accurately screen for sanctions and compliance risks, ensuring explainability, proportionality, and minimising false positives in high-stakes environments.	Evaluates whether AI tools in compliance contexts meet legal standards for accuracy, fairness, and accountability. Ensures proper safeguards are in place to prevent discriminatory or disproportionate outcomes.	Are subject to decisions based on automated screening. Should understand how such AI systems operate and have access to explanations and appeal mechanisms in case of adverse outcomes.
13. AI and Anti-Money Laundering (AML)	Reviews AI for risk scoring, transaction monitoring, and red flag detection under AMLR, TFR, and GDPR.	Must ensure AI systems used for AML comply with legal standards for risk scoring, transaction monitoring, and anomaly detection, while balancing accuracy with privacy and fairness.	Oversees AI deployment in AML to ensure effectiveness, proportionality, and adherence to GDPR and AMLR. Evaluates transparency, auditability, and the minimisation of discriminatory impacts.	May be affected by AI systems that assess financial risk or detect suspicious activity. Should be informed how these systems evaluate transactions, what triggers alerts, and have the right to challenge or clarify decisions that impact them.





14. Digital Operational Resilience (DORA) for AI in Finance	Assesses how AI used in financial services is regulated under DORA, including testing, resilience, and third-party risk.	Must ensure AI systems in financial services meet DORA requirements, including robust testing, operational resilience, incident response, and third-party risk management.	Supervises AI systems under DORA to ensure they can withstand operational disruptions. Evaluates resilience measures, risk controls, and compliance with ICT-related obligations in financial institutions.	Rely on stable and secure AI-driven financial services. Should be protected from disruptions and informed about safeguards in place to ensure system reliability and continuity.
15. Classification of AI Systems Under the AI Act	Provides a step-by-step framework for classifying AI systems (e.g. prohibited, high-risk, limited-risk) under the AI Act.	Must assess and classify their AI systems using the AI Act's framework (prohibited, high-risk, limited-risk, minimal-risk) to determine applicable obligations, including conformity assessments and transparency requirements.	Verifies that AI systems are correctly classified and that appropriate compliance measures are applied. Provides oversight and guidance, particularly for high-risk and borderline cases.	Should be informed of the risk level of AI systems they interact with. This enables them to understand the system's purpose, limitations, and their rights under the AI Act.
16. Real-World Testing & Sandbox Exemptions (AI Act Articles 57–59)	Details the sandbox's regulatory testing framework, including conditions for exemptions and supervisory oversight.	Can leverage sandbox exemptions for real-world testing of AI systems, allowing for regulatory flexibility while ensuring safety, transparency, and compliance with the AI Act.	Oversees the sandbox process, ensuring that AI systems are tested in controlled environments. Monitors compliance with AI Act conditions, including risk mitigation and supervisory oversight during testing.	Should know that AI products tested in the sandbox follow specific rules and oversight. This helps ensure safety, transparency, and fairness before full market release, and users should be informed of any limitations or exemptions applied during testing.
17. Open Source AI – Roles & Responsibilities	Explores challenges in assigning responsibility for open-source AI systems under the AI Act.	Must define clear roles and responsibilities when developing or deploying open-source AI, ensuring compliance with the AI Act, transparency in model usage, and	Monitors open-source AI systems to ensure they meet regulatory requirements. Provides guidance on liability, risk management, and transparency, while addressing	Rely on open-source AI for transparency and innovation. Should be aware of the potential risks, including lack of formal oversight, and have access to information on the system's





		addressing accountability for potential risks or harms.	challenges in accountability for distributed development.	development, testing, and safety standards.
18. AI for Crowdfunding, Finance, and Tokenized Assets	Evaluates AI in financial applications, especially for credit scoring, profiling, and alignment with MiFID and ECSPR.	Must ensure AI systems used in crowdfunding, financial services, and tokenized assets comply with regulations like MiFID and ECSPR, providing accurate profiling, risk assessment, and transparent decision-making.	Monitors AI applications in finance and crowdfunding to ensure compliance with financial laws and regulations. Ensures that AI-driven tools for asset management and tokenization are transparent, fair, and secure.	Benefit from AI-enhanced financial services, with clear insights into risk, investment opportunities, and asset value. Should be informed of how AI influences financial decisions and be provided with safeguards against unfair treatment or biased algorithms.
19. Automated Decision-Making & Algorithmic Markets	Looks at governance of AI-powered agents and algorithmic trading systems in decentralised or autonomous settings.	Must ensure that automated decision-making systems in markets are transparent, fair, and comply with legal standards, including safeguarding against manipulation and ensuring accountability in algorithmic trading.	Oversees AI systems in financial markets to ensure compliance with regulations governing automated trading, fairness, and transparency. Monitors algorithmic decisions for risks like market manipulation or systemic failure.	Should be informed about how AI systems make decisions in trading and market activities. They should have access to transparent explanations and redress mechanisms if adversely affected by automated decisions.



Calls to action





Plan Early for Sandbox Participation

Engaging in a regulatory sandbox requires clear objectives, documented test plans, and defined risk mitigation strategies. Early preparation enables AI firms to make the most of sandbox exemptions, reduce compliance friction, and gather actionable insights from regulators during real-world testing.



Align Sandbox Testing with AI Act Risk Classes

Sandbox trials should be structured around the AI Act's risk classification framework—especially for high-risk systems under Title III. Tailoring testing protocols to specific obligations (e.g., data governance, human oversight) ensures that results translate directly into conformity assessment readiness.



Foster Regulator Collaboration Across Jurisdictions

Cross-border deployment of AI systems demands coordinated regulatory input. Use the sandbox to build trusted relationships with national and EU authorities, harmonize interpretations, and address overlapping rules such as GDPR, eIDAS, or sectoral laws in a single engagement.



Incorporate User-Centric Feedback Loops

Cross-border deployment of AI systems demands coordinated regulatory input. Use the sandbox to build trusted relationships with national and EU authorities, harmonize interpretations, and address overlapping rules such as GDPR, eIDAS, or sectoral laws in a single engagement.





Conclusion

The introduction of AI regulatory sandboxes under the EU AI Act represents a transformative shift in how high-risk AI systems are developed, tested, and governed—moving toward a more collaborative, transparent, and risk-based compliance model. In offering a supervised environment for real-world experimentation, sandboxes help translate the AI Act’s legal obligations into practical implementation, reducing uncertainty for innovators while reinforcing user protections and regulatory accountability.

Throughout this report, we have highlighted how sandboxes serve as a convergence point for AI firms, regulators, and users—facilitating early feedback, technical oversight, and safer, more explainable AI systems. When thoughtfully designed, these environments can accelerate alignment with the AI Act’s requirements for data governance, human oversight, and lifecycle risk management, while also fostering trust and innovation across the EU AI ecosystem.

However, successful deployment of regulatory sandboxes depends on more than good policy design—it requires operational readiness and execution. Stakeholders must adapt internal compliance processes, documentation practices, and governance models to reflect the AI Act’s expectations, particularly for high-risk use cases defined under Annex III. Smaller developers and first-time participants may face steeper learning curves, making early planning, regulatory guidance, and cross-functional collaboration essential.

Despite these complexities, forward-leaning organizations and authorities are already showing that AI sandboxes can streamline compliance, improve system quality, and reduce time to market. As AI becomes increasingly embedded across sectors—from healthcare and finance to public administration and civic infrastructure—regulatory sandboxes provide a powerful tool for building systems that are not only innovative, but also lawful, ethical, and aligned with fundamental rights. To conclude, Successful sandbox outcomes could shape future AI Act revisions and set a global benchmark for AI governance.





Appendix A: A Practitioner's View: Navigating the 'Triple Bind' of Multi- Regulatory Compliance in AI Sandboxes for Financial Services



Michael Godfriaux

Euroclear



For financial institutions looking to innovate with high-risk AI, the primary challenge is not merely complying with the EU AI Act in isolation. Rather, it is navigating the simultaneous, and at times conflicting, demands of three distinct regulatory pillars: the AI Act itself, sector-specific mandates like the Digital Operational Resilience Act (DORA), and horizontal laws such as GDPR and the EU's Anti-Money Laundering Regulation (AMLR). This creates a "Triple Bind" that AI Regulatory Sandboxes (AIRS) are uniquely positioned to address.

The AIRS report correctly identifies the need for handling multi-regulatory compliance by learning from blockchain sandbox precedents. The challenge can be broken down as follows:

1. **The AI Act:** For high-risk systems common in finance (e.g., credit scoring, insurance pricing, AML transaction monitoring), the Act mandates stringent transparency, detailed technical documentation, human oversight, and explainability.
2. **DORA (Digital Operational Resilience Act):** This regulation treats AI systems that support critical business functions as vital ICT components. It imposes rigorous obligations for risk management, resilience testing, and incident reporting, introducing a focus on system stability and continuity that can be in tension with the experimental nature of a sandbox environment.



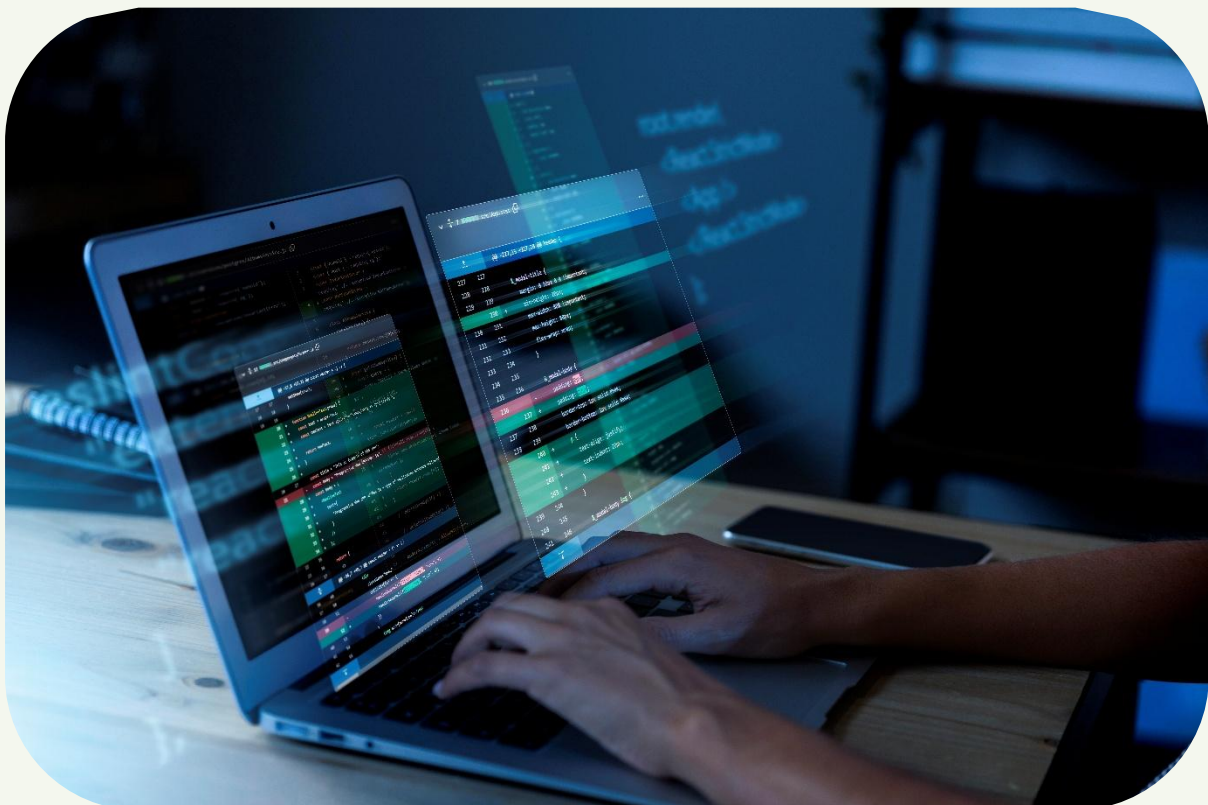


3. **GDPR & AML Regulation:** These frameworks impose strict rules on the processing of personal data, the right to an explanation for automated decisions (Article 22 GDPR), and the critical need for confidentiality in suspicious activity reporting (AMLR). This principle of confidentiality can directly conflict with the AI Act's push for radical transparency.

An illustrative use case is the testing of an AI-powered AML monitoring system within a sandbox. The firm faces a difficult set of questions: How can it provide full transparency on its AI model's logic to satisfy the AI Act without revealing sensitive detection rules that could be exploited by criminals, thereby violating AML principles? How does it test the system with "real-world" data to meet the AI Act's testing requirements while adhering to GDPR's principles of data minimization and purpose limitation?

The solution lies in evolving sandboxes from purely technological testbeds into platforms for *regulatory de-confliction*. We propose the establishment of a **"Joint Regulatory Guidance Protocol"** within sandboxes for the financial sector. This protocol would facilitate the simultaneous participation and guidance of the relevant Data Protection Authority (DPA), the National Competent Authority for finance (e.g., Luxembourg's CSSF), and the national Financial Intelligence Unit (FIU). This approach operationalizes the report's excellent recommendations for effective dialogue structures and pan-European coordination.

Ultimately, the success of AI sandboxes will hinge on their ability to provide a forum where innovators and a triumvirate of regulators can collaboratively resolve these conflicts, ensuring that innovation proceeds not in spite of regulation, but in concert with it.





Appendix B: A Practitioner's View: Rights-Respecting Approach to Regulation



Nat. C. Schooler

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From my experience as an IBM AI Futurist, I believe that now the time is ripe to rollout AI with caution. And I am certain you agree sandboxes offer a good way to remain safe. Avoiding fines for non-compliance is certainly of preference. Robust system design and integration, business led innovation and ethical considerations are paramount. And I am sure you agree sandboxes offer a structured, controlled environment to test and validate innovative AI systems under regulatory oversight. Sandboxes are a game-changer as they have been in the past! They enable early feedback, legal clarity, and iterative improvement, significantly accelerating compliance with the EU AI Act. Those are crucial for de-risking innovation and ensuring that AI solutions can reach the market responsibly and adhere to regulations.

However, as you know the "air of caution" is equally important. Even when implementing AI agents and other tools. The success of these sandboxes hinges on overcoming several practical hurdles. One key lesson from blockchain sandboxes, is the absolute necessity of a harmonized terminology and clear legal definitions. Unclear or inconsistent use of terms like "AI system" or "high-risk" can truly impede effective regulatory dialogue and create unnecessary confusion. The importance of addressing multi-regulatory compliance is a given. Navigating overlaps with other legislation such as GDPR, eIDAS, and the Digital Services Act (DSA) are crucial. My concern is that without robust tools like modular legal mapping and joint regulator engagement, innovators could receive inconsistent advice, slowing progress. The focus on a "human-centric approach" continues to be vital for non-bias. Incorporating user-centric feedback loops ensures that AI systems are developed with fairness and accountability from the ground up, not just as a post-deployment fix.

In conclusion, these AI Regulatory Sandboxes are more than just regulatory mechanisms. They represent a collaborative, rights-respecting approach to AI experimentation. They are a powerful tool for accelerating compliance, refining oversight, and ultimately, building trustworthy and explainable AI systems before they reach the broader market. My advice is to embrace these sandboxes as a strategic asset, map your innovation journey and all the moving parts. Plan early for participation, align testing with the AI Act's risk classes, foster cross-jurisdictional regulator collaboration, and critically, incorporate user-centric feedback. This forward-leaning engagement isn't just about avoiding penalties. It is about leading with integrity, shaping the future of AI and reaping the rewards.





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